

SARDAR PATEL UNIVERSITY
M. Sc. (IT) INTEGRATED EXAMINATION, VIII SEM

Date : 17th APR 2017, Monday
 Session : Evening
 Course No : PS08CIIT03

Time : 02:00pm To 05:00pm
 Sub: Artificial Intelligence
 Total marks : 70

Q – 1 Select an appropriate option

[08]

- i) Common sense reasoning is a _____.
 a) Mundane Task b) Formal Task c) Expert Task d) None
- ii) What is Artificial intelligence?
 a) Making a machine intelligent
 b) Programming with your own intelligence
 c) Putting your intelligence into computer
 d) Putting your memory into computer
- iii) Factual knowledge uses _____ symbol for 'OR'.
 a) $\wedge$ b) $\vee$ c) $\sim$ d) $\implies$
- iv) _____ is the primary consideration in designing knowledge based AI.
 a) Analogy b) Efficacy c) Efficiency d) Acquisition
- v) _____ maps elements of a fuzzy set to real numbered values in the interval 0 to 1.
 a) Fuzzy Logic b) Bi-valued Logic c) Membership Function d) None
- vi) In fuzzy connectives implications are obtained by _____.
 a) Union b) Intersection c) Negation & Intersection d) Negation
- vii) _____ is a collection of such genes representing possible solutions in domain space.
 a) Genetic Cycle b) Genetic algorithm c) Mutation d) Genotype
- viii) The principle of Genetic Algorithm is based on _____.
 a) Natural Evolution (Survival of fittest) b) Blind Search
 c) Stored program concept d) None of these

Q – 2 Answer the following Questions (Any SEVEN)

[14]

- i) Explain information and knowledge.
- ii) Write in brief about components of KBS.
- iii) List four reasons for solving complex problem using AI technique.
- iv) List Difficulties in KBS Development explain any one.
- v) Explain concept of Updating Knowledge.
- vi) Explain the limitations of Fuzzy System.
- vii) Explain Integrating Neural Networks and Knowledge Based System.
- viii) Write Genetic Algorithm steps.
- ix) What do you mean by genotype and fitness?

- Q – 3 a) Explain the types of Knowledge. [06]
 b) Explain Limitations of Knowledge Based System. [06]

OR

- Q – 3 b) Write note on inference engine. [06]

- Q – 4 (a) Explain Factual Knowledge. [06]
 (b) Explain Knowledge based system tools. [06]
- OR**
- Q – 4 (b) Explain knowledge acquisition. [06]
- Q – 5 a) Explain operations of Fuzzy sets and types of Fuzzy functions. [06]
 b) Explain Fuzzy control system in detail. [06]
- OR**
- Q – 5 b) Write Learning paradigms of neural network. [06]
- Q – 6 a) Explain basic operators of a Genetic Algorithm [06]
 b) Write about application and research trends in Genetic Algorithm. [06]
- OR**
- Q – 6 b) Minimize a simple function $f(x, y) = x + y$ on the integer interval $[0,30]$ using genetic algorithm. The encoding strategy for the example is to follow binary bit pattern of size 10, with the first 5 bits representing a value of the first variable x and remaining bit representing a y . Take the 01101, 11000, 01000, 10011 strings as individuals in the first population x and 01010, 01100, 00101, 00111 strings as individuals in the second population y . [06]

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(2)